Primovist® 0.25 mmol/ml

Solution for injection

1. NAME OF THE MEDICINAL PRODUCT

Primovist 0.25 mmol/ml, solution for injection.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each ml contains 0.25 mmol gadoxetate disodium (equivalent to 181.43 mg gadoxetate disodium).
For full list of excipients, see section “List of excipients“.

3. PHARMACEUTICAL FORM

Solution for injection.

4. CLINICAL PARTICULARS

4.1 Indication(s)

This medicinal product is for diagnostic use only.

Primovist is indicated for use in adults for the enhancement of magnetic resonance imaging (MRI) of focal liver lesions.

Primovist should be used only when diagnostic information is essential and not available with unenhanced magnetic resonance imaging (MRI) and when delayed phase imaging is required.

4.2 Dosage and method of administration

4.2.1 Method of administration

This medicinal product is for intravenous administration.

The dose is administered undiluted as a bolus injection. After the injection of the contrast medium the intravenous cannula/line should be flushed using physiological saline solution.

After bolus injection of Primovist, dynamic imaging during arterial, portovenous, and equilibrium phases utilizes the different temporal enhancement pattern of different liver lesion types to obtain information about their classification (benign/malignant) and the specific characterization. It further improves visualization of hypervascular liver lesions.

The delayed (hepatocyte) phase starts at about 10 minutes post injection (in confirmatory studies most of the data were obtained at 20 minutes post injection) with an imaging window lasting at least 120 minutes. The imaging window is reduced to 60 minutes in patients requiring hemodialysis and in patients with elevated bilirubin values (> 3 mg/dl) (see also section ‘Interaction with other medicinal products and other forms of interaction’).

For additional instructions see section “Instructions for use/handling”.

4.2.2 Dosage regimen

The lowest dose that provides sufficient enhancement for diagnostic purposes should be used. The dose should be calculated based on the patient’s body weight, and should not exceed the recommended dose per kilogram of body weight detailed in this section.

Adults:
0.1 ml per kg body weight Primovist (equivalent to 25 µmol per kg body weight)

4.2.3 Additional information on special populations

Pediatric population
Primovist is not recommended for use in children below 18 years of age due to a lack of data on safety and efficacy.

Patients with renal impairment
In clinical studies, no overall differences in safety and efficacy were observed between patients with renal impairment and patients with normal kidney function. The elimination of gadoxetate disodium is prolonged in renally impaired patients. To ensure diagnostically useful images, no dosage adjustment is recommended (see also section ‘Special warnings and precautions for use’).

4.3 Contraindications

None

4.4 Special warnings and special precautions for use

Exposure to gadolinium-based contrast agents (GBCAs) increases the risk for Nephrogenic Systemic Fibrosis (NSF) in patients with:
► acute or chronic severe renal insufficiency (glomerular filtration rate < 30mL/min/1.73m²), or
► acute renal insuficiency of any severity due to the hepato-renal syndrome or in the perioperative liver transplantation period.
NSF is a debilitating and sometimes fatal disease affecting the skin, muscle, and internal organs.
Avoid use of GBCAs unless the diagnostic information is essential and not available with non-contrast enhanced magnetic resonance imaging (MRI).
Screen all patients for renal dysfunction by obtaining a history and/or laboratory tests.
When administering a GBCA, do not exceed the dose recommended in product labelling. Allow sufficient time for elimination of the GBCA prior to any readministration.

- Among the factors that may increase the risk for NSF are repeated or higher than recommended doses of a GBCA.
- For patients receiving hemodialysis, healthcare professionals may consider prompt hemodialysis following GBCA administration in order to enhance the contrast agent’s elimination. However, it is unknown if hemodialysis prevents NSF.
- Determine the renal function of patients by obtaining a medical history of conducting laboratory tests that measure renal function prior to using GBCA.
- The risk, if any, for developing NSF among patients with mild to moderate renal insufficiency or normal renal function is unknown.
- Post-marketing reports have identified the development of NSF following single and multiple administrations of GBCAs.

► Risks associated with intrathecal use
Serious, life-threatening and fatal cases, primarily with neurological reactions (e.g. coma, encephalopathy, seizures), have been reported with intrathecal use of gadolinium-based contrast agents (GBCAs). The safety and effectiveness of Primovist have not been established with intrathecal use. Primovist is not approved for intrathecal use

► Hypersensitivity
Particularly careful risk-benefit assessment is required in patients with known hypersensitivity to Primovist.

As with other intravenous contrast agents, Primovist can be associated with anaphylactoid/hypersensitivity or other idiosynchratic reactions characterized by cardiovascular, respiratory and cutaneous manifestations, and ranging to severe reactions including shock.

The risk of hypersensitivity reactions is higher in case of:
- previous reaction to contrast media
- history of bronchial asthma
- history of allergic disorders.

In patients with an allergic disposition the decision to use Primovist must be made after particularly careful evaluation of the risk-benefit ratio.

Most of these reactions occur within half an hour of administration. Therefore, post-procedure observation of the patient is recommended. Medication for the treatment of hypersensitivity reactions as well as preparedness for institution of emergency measures are necessary.

Delayed reactions after hours up to several days have been rarely observed (see section “Undesirable effects”).

► Cardiovascular disease
Caution should be exercised when Primovist is administered to patients with severe cardiovascular problems because only limited data are available so far.

► Impaired renal function
In healthy subjects, gadoxetate disodium is equally eliminated via renal and hepatobiliary routes.

Prior to administration of Primovist, it is recommended, that all patients are screened for renal dysfunction by obtaining a history and/or laboratory tests.

In patients with severely impaired renal function, the benefits must be weighed carefully against the risks, since contrast medium elimination is delayed in such cases. A sufficient period of time for elimination of the contrast agent from the body prior to any re-administration in patients with renal impairment should be ensured.

Gadoxetate disodium can be removed from the body by hemodialysis. About 30% of the administered dose is eliminated from the body by a single dialysis session of 3 hours starting 1 hour post injection. In end-stage renal failure patients, gadoxetate disodium was almost completely eliminated via dialysis and biliary excretion within the observation period of 6 days, the majority within 3 days.

For patients already receiving hemodialysis at the time of Primovist administration, prompt initiation of hemodialysis following the administration of Primovist should be considered, in order to enhance the contrast agent's elimination.

There have been reports of nephrogenic systemic fibrosis (NSF) associated with the use of some contrast agents containing gadolinium in patients with
- acute or chronic severe renal impairment (GFR < 30 ml/min/1.73m^2) and
- acute renal insufficiency of any severity due to the hepato-renal syndrome or in the perioperative liver transplantation period.

Although the systemic body exposure with gadolinium is low based on the diagnostic dosage of Primovist as well as its dual elimination pathways (renal and hepatobiliary), there is a possibility that NSF may occur with Primovist. Therefore, Primovist should only be used in these patients after careful risk/benefit assessment (see section “Undesirable effects”).

► Local intolerance
Intramuscular administration must be strictly avoided, because it may cause local intolerance reactions including focal necrosis (see section ”Preclinical safety data”).

► Accumulation in the body
After administration of gadoxetate disodium gadolinium can be retained in the brain and in other tissues of the body (bones, liver, kidneys, skin) and can cause dose-dependent increases in T1-weighted signal intensity in the brain, particularly in the dentate nucleus, globus pallidus, and thalamus. Clinical consequences are unknown. The possible diagnostic advantages of using gadoxetate disodium in patients who will require repeated scans should be weighed against the potential for deposition of gadolinium in the brain and other tissues.

4.5 Interaction with other medicinal products and other forms of interaction

Interference with OATP inhibitors
Animal studies demonstrated that compounds belonging to the class of anionic medicinal products, e.g. rifampicin, block the hepatic uptake of Primovist thus reducing the hepatic contrast effect. In this case, the expected benefit of an injection of Primovist might be limited. No other interactions with medicinal products are known from animal studies.

An interaction study in healthy subjects demonstrated that the co-administration of the OATP inhibitor erythromycin did not influence efficacy and pharmacokinetics of Primovist. No further clinical interaction studies with other medicinal products have been performed.

Interference from elevated bilirubin or ferritin levels in patients
Elevated levels of bilirubin (>3 mg/dl) or ferritin can reduce the hepatic contrast effect of Primovist. If Primovist is used in these patients, complete the magnetic resonance imaging no later than 60 minutes after Primovist administration.

Interference with diagnostic tests
Serum iron determination using complexometric methods (e.g. Ferrocine complexation method) may result in falsely high or low values for up to 24 hours after the examination with Primovist because of the free complexing agent caloxetate trisodium contained in the contrast medium solution.

4.6 Pregnancy and lactation

4.6.1 Pregnancy
Traces of gadolinium-based contrast agents can cross the placental barrier and lead to fetal exposure.
The potential risks of an abnormal pregnancy outcome are unknown, as adequate and well controlled clinical studies with Primovist were not conducted in pregnant women.
A retrospective cohort study comparing pregnant women who had a GBCA MRI to pregnant women who did not have an MRI reported a higher occurrence of stillbirths and neonatal deaths in the group receiving GBCA MRI. However, no increased risk of congenital anomalies was observed. Limitations of this study include a lack of comparison with non-contrast MRI and lack of information about the maternal indication for MRI.

These limitations were further addressed in another retrospective cohort study that found no increased risk for fetal or neonatal death or Neonatal Intensive Care Unit (NICU) admission when comparing pregnancies exposed to GBCA MRI and non-contrast MRI.
Cohort studies and case reports on exposure to GBCAs during pregnancy have not reported a clear association between GBCAs and congenital anomalies, fetal and neonatal death or the need for neonatal intensive care unit admission.
Animal studies at clinically relevant doses have not shown reproductive toxicity after repeated administration (see section ‘*5.3 Preclinical safety data*’).
Primovist should only be used during pregnancy after a careful benefit-risk analysis.

4.6.2 Lactation

It is unknown whether gadoxetate disodium is excreted in human milk.

There is evidence from non-clinical data that gadoxetate is excreted into breast milk in very small amounts (less than 0.5% of the dose intravenously administered) and the absorption via the gastrointestinal tract is poor (about 0.4 % of the dose orally administered were excreted in the urine).

At clinical doses, no effects on the infant are anticipated and Primovist can be used during breastfeeding.

4.7 Effects on ability to drive or use machines

Not known.

4.8 Undesirable effects

4.8.1 Summary of the safety profile

The overall safety profile of Primovist is based on data from more than 1,900 patients in clinical trials, and from post-marketing surveillance.

The most frequently observed adverse drug reactions (≥ 0.5 %) in patients receiving Primovist are nausea, headache, feeling hot, blood pressure increased and dizziness.

The most serious adverse drug reaction in patients receiving Primovist is anaphylactoid shock.
Delayed allergoid reactions (hours later up to several days) have been rarely observed.

Most of the undesirable effects were of mild to moderate intensity.

4.8.2 Tabulated list of adverse reactions

The adverse drug reactions observed with Primovist are represented in the table below. They are classified according to System Organ Class. The most appropriate MedDRA term is used to describe a certain reaction and its synonyms and related conditions.

Adverse drug reactions from clinical trials are classified according to their frequencies. Frequency groupings are defined according to the following convention: common: ≥ 1/100 to < 1/10; uncommon: ≥ 1/1,000 to < 1/100; rare: ≥ 1/10,000 to < 1/1,000. The adverse drug reactions identified only during post-marketing surveillance, and for which a frequency could not be estimated, are listed under ‘not known’.

Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

Table 1: Adverse drug reactions reported in clinical trials or during post-marketing surveillance in patients treated with Primovist

System Organ Class (MedDRA)	Common	Uncommon	Rare	Not known
Immune system disorders				Hypersensitivity /anaphylactoid reaction (e.g. shock*, hypotension, pharyngolaryngeal edema, urticaria, face edema, rhinitis, conjunctivitis, abdominal pain, hypoesthesia, sneezing, cough, pallor)
Nervous system disorders	Headache	Vertigo Dizziness Dysgeusia Paresthesia Parosmia	Tremor Akathisia	Restlessness
Cardiac disorders			Bundle branch block Palpitation	Tachycardia

System Organ Class (MedDRA)	Common	Uncommon	Rare	Not known
Vascular disorders		Blood pressure increased Flushing		
Respiratory, thoracic and mediastinal disorders		Respiratory disorders (Dyspnea*, Respiratory distress)		
Gastrointestinal disorders	Nausea	Vomiting Dry mouth	Oral discomfort Salivary hypersecretion	
Skin and subcutaneous tissue disorders		Rash Pruritus**	Maculopapular rash Hyperhidrosis	
Musculoskeletal, connective tissue and bone disorders		Back pain		
General disorders and administration site conditions		Chest pain Injection site reaction*** Feeling hot Chills Fatigue Feeling abnormal	Discomfort Malaise	

* Life-threatening and/or fatal cases have been reported. These reports originated from post-marketing experience.

** Pruritus (Generalized pruritus, Eye pruritus)

*** Injection site reactions (various kinds) comprise the following terms: Injection site extravasation, Injection site burning, Injection site coldness, Injection site irritation, Injection site pain

4.8.3 Description of selected adverse reactions

Cases of nephrogenic systemic fibrosis (NSF) have been reported with some contrast agents containing gadolinium (see also ‘Special warnings and precautions for use’).

Slightly elevated serum iron and serum bilirubin values have been observed in less than 1% of patients after administration of Primovist. However, the values did not exceed more than 2–3 times the baseline values and returned to their initial values without any symptoms within 1 to 4 days.

4.9 Overdose

Single doses of gadoxetate disodium as high as 0.4 ml/kg (100 µmol/kg) body weight were tolerated well. There have been no cases of overdose observed or reported in clinical use. Therefore, the signs and symptoms of overdosage have not been characterized.

► Patients with renal and/or hepatic impairment

In case of inadvertent overdosage in patients with severely impaired renal and/or hepatic function, Primovist can be removed from the body by hemodialysis (see section ‘Special warnings and precautions for use’).

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: paramagnetic contrast media
ATC Code: V08C A10

Mechanism of action

Primovist is a paramagnetic contrast agent for magnetic resonance imaging.

The contrast-enhancing effect is mediated by gadoxetate, an ionic complex consisting of gadolinium (III) and the ligand ethoxybenzyl-diethylenetriamine-pentaacetic acid (EOB-DTPA).

When T_1-weighted scanning sequences are used in proton magnetic resonance imaging, the gadolinium ion-induced shortening of the spin-lattice relaxation time of excited atomic nuclei leads to an increase of the signal intensity and, hence, to an increase of the image contrast of certain tissues.

Pharmacodynamic effects
Gadoxetate disodium leads to a distinct shortening of the relaxation times even at low concentrations. At pH 7, a magnetic field strength of 0.47 T and 40°C the relaxivity (r_1) - determined from the influence on the spin lattice relaxation time (T_1) of protons in plasma is about 8.18 l/(mmol.sec) and the relaxivity (r_2) - determined from the influence on the spin-spin relaxation time (T_2)- is about 8.56 l/(mmol*sec). At 1.5 T and 37°C the respective relaxivities in plasma are r_1 = 6.9 l/(mmol.sec) and r_2 = 8.7 l/(mmol.sec). The relaxivity displays a slight inverse dependency on the strength of the magnetic field.

Ethoxybenzyl-diethylenetriaminepentaacetate forms a stable complex with the paramagnetic gadolinium ion with extremely high in-vivo and in-vitro stability (thermodynamic stability constant: log K_{GdI} = 23.46). Gadoxetate disodium is a highly water-soluble, hydrophilic compound with a partition coefficient between n butanol and buffer at pH 7.6 of about 0.011.

Due to its lipophilic ethoxybenzyl moiety gadoxetate disodium exhibits a biphasic mode of action: first, distribution in the extracellular space after bolus injection and subsequently selective uptake by hepatocytes. The relaxivity r_1 in liver tissue is 16.6 l/(mmol.sec) (at 0.47T) resulting in increased signal intensity of liver tissue. Subsequently gadoxetate disodium is excreted into the bile.

The substance does not display any significant inhibitory interaction with enzymes at clinically relevant concentrations.

5.2 Pharmacokinetic properties

General introduction
Gadoxetate disodium behaves in the organism like other highly hydrophilic biologically inert, renally and hepatobiliary excreted compounds.

Absorption and Distribution
After intravenous administration, the plasma concentration time profile of gadoxetate disodium is characterized by a bi-exponential decline. The total distribution volume of gadoxetate disodium at steady state is about 0.21 l/kg (extracellular space). The plasma protein binding is less than 10%.

The compound diffuses through the placental barrier only to a small extent as demonstrated in rats.

In lactating rats, less than 0.5% of the intravenously administered dose (0.1 mmol/kg) of radioactively labeled gadoxetate was excreted into the breast milk. Absorption after oral administration was very small in rats with 0.4%.

Gadoxetate disodium is a linear GdCA. Studies have shown that after exposure to GdCAs gadolinium is retained in the body. This includes retention in the brain and in other tissues and organs. With the linear GdCAs this can cause dose-dependent increases in T1-weighted signal intensity in the brain, particularly in the dentate nucleus, globus pallidus, and thalamus. Signal intensity increases and non-clinical data show that gadolinium is released from linear GdCAs.

Metabolism
Gadoxetate disodium is not metabolized.

Elimination
Gadoxetate disodium is completely excreted in equal amounts via the renal and hepatobiliary routes.

Seven days after intravenous injection of gadoxetate, less than 1% of the dose administered was found in the bodies of rats and monkeys. Of this, the highest concentration was found in kidney and liver.

The mean terminal elimination half–life of gadoxetate disodium (dose 0.01 to 0.1 mmol/kg) observed in healthy subjects was about 1 hour.

The total serum clearance (CL) was 250 ml/min. The renal clearance (CL_R) corresponds to about 120 ml/min, a value similar to the glomerular filtration rate in healthy subjects.

Linearity/non-linearity
Gadoxetate disodium shows linear pharmacokinetics i.e. pharmacokinetic parameters change dose proportionally (e.g. C_{max}, AUC) or are dose independent (e.g. V_{ss}, $t_{1/2}$), up to a dose of 100 µmol/kg body weight (0.4 ml/kg).

Characteristics in special patient populations
A phase III study with 25 µmol per kg body weight Primovist compared subjects with various levels of impaired hepatic function, impaired renal function, coexistent hepatic and renal impairment, and healthy subjects of different age groups, including elderly.

► Gender
Total clearance was about 20% lower in female (185 ml/min) than in male subjects (236 ml/min).

► Elderly population (aged 65 years and above)
In accordance with the physiological changes in renal function with age, the plasma clearance of gadoxetate disodium was reduced from 210 ml/min in non-elderly subjects to 163 ml/min in elderly subjects aged 65 years and above. Terminal half-life and systemic exposure were higher in the elderly (2.3 h and 197 μmol*h/l, respectively) compared to the control group (1.8 h and 160 μmol*h/l, respectively). The renal excretion was complete after 24 h in all subjects with no difference between elderly and non-elderly healthy subjects.

► Renal and/or hepatic impairment
In patients with moderate renal impairment, an increase in AUC to 237 μmol*h/l and of terminal half-life to 2.2 h was observed. In patients with end-stage renal failure, the AUC was increased to about 903 μmol*h/l and the terminal half-life prolonged to about 20 h in patients. About 55% of the administered dose was recovered in feces within the observation period of 6 days, the majority within 3 days.

In patients with mild or moderate hepatic impairment, a slight to moderate increase in plasma AUC, half-life and urinary excretion, as well as a decrease in hepatobiliary excretion were observed in comparison to healthy subjects.

In patients with severe hepatic impairment, especially in patients with abnormally high serum bilirubin levels (> 3 mg/dl), the AUC was increased to 259 μmol*h/l compared to 160 μmol*h/l in the control group. The elimination half-life was increased to 2.6 h compared to 1.8 h in the control group. The hepatobiliary excretion substantially decreased to 5.7% of the administered dose in these patients.

Gadoxetate disodium can be removed from the body by hemodialysis. About 30 % of the administered dose were recovered in the dialysate in a 3 hour dialysis starting 1 hour post injection. In the study with end-stage renal failure patients, gadoxetate disodium was almost completely eliminated via dialysis and biliary excretion within 6 days. Plasma concentrations of gadoxetate disodium were measurable up to 72 hours post-dose in these patients (see section 'Special warnings and precautions for use').

5.3 Preclinical safety data

Non-clinical data reveal no special hazard for humans based on conventional studies of systemic toxicity, genotoxicity, contact-sensitizing potential.

► Reproduction toxicology
Repeated intravenous dosing of Primovist in studies on embryofetal development caused embryotoxicity (increased post implantational loss) in rabbits at 25.9 times (based on body surface area) or 80 times (based on body weight) the human single dose.

► Local tolerance
Experimental local tolerance studies with Primovist indicated good local tolerability after intravascular (intravenous and intraarterial) and paravenous administration.

However, intramuscular administration caused local intolerance reactions and must therefore be strictly avoided in humans (see section 'Special warnings and precautions for use').

6. PHARMACEUTICAL PARTICULARS

List of excipients
Caloxetate trisodium
Hydrochloric acid (for pH adjustment)
Sodium hydroxide (for pH adjustment)
Trometamol
Water for injection

Incompatibilities
In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

Shelf life
Please refer to labels.

► Shelf life after first opening of the container
Primovist should be used immediately after opening.

Nature and contents of container
1 x 10 ml (in 10 ml prefilled syringe)

Instructions for use / handling
Visual Inspection
This medicinal product should be visually inspected before use.

Primovist should not be used in case of severe discoloration, the occurrence of particulate matter or a defective container.

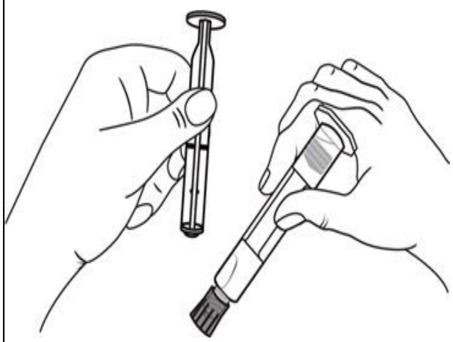
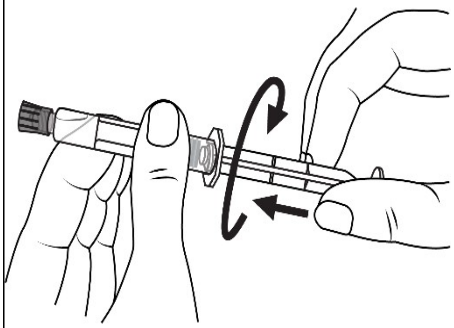
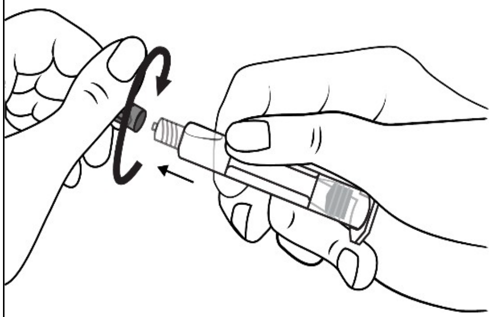
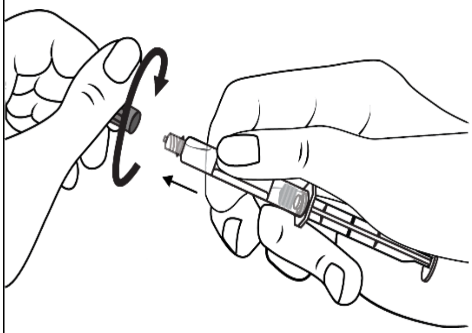
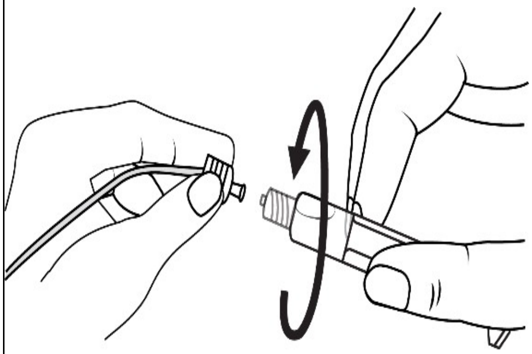
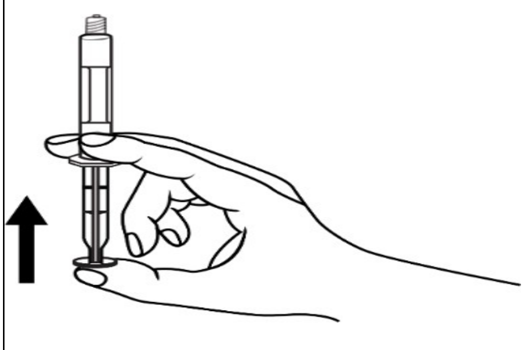
Prefilled syringes
The prefilled syringe should be prepared for the injection immediately before the examination.

The tip cap should be removed from the prefilled syringe immediately before use.

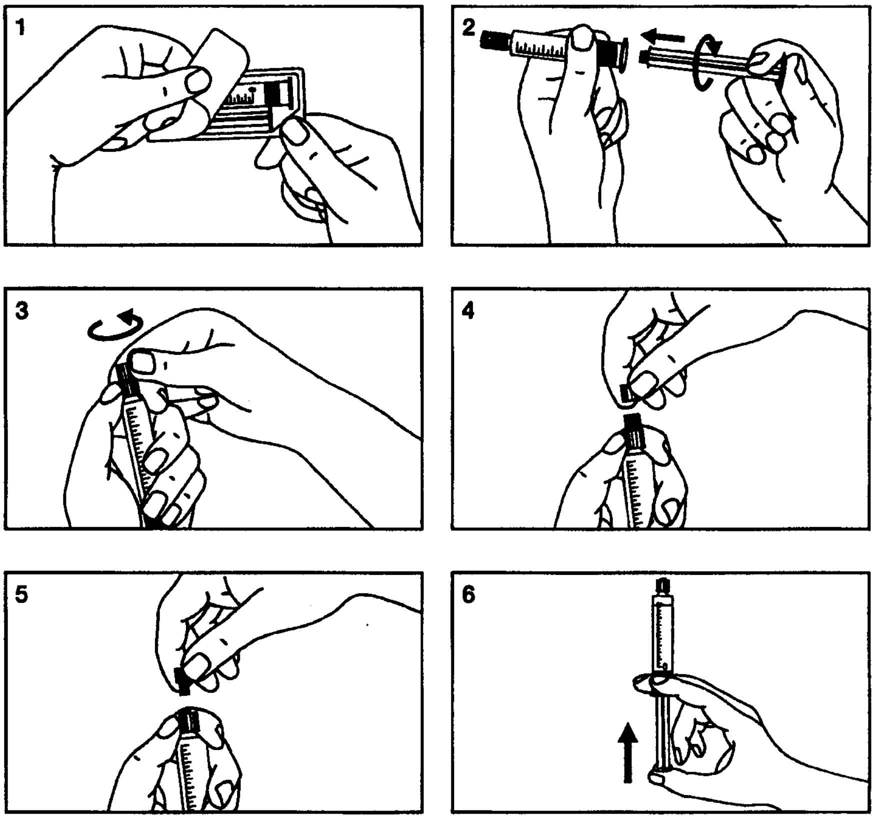
Any solution not used in one examination is to be discarded in accordance with local requirements.

The peel-off tracking label on the syringes should be stuck onto the patient record to enable accurate recording of the gadolinium contrast agent used. The dose used should also be recorded. If electronic patient records are used, the name of the product, the batch number and the dose should be entered into the patient record.

Plastic syringe only:

Hand injection	Injection with a power injector
 1. Take out the syringe and plunger rod	 1. Take out the syringe
 2. Turn the plunger rod clockwise into the syringe	 2. Open the cap with a twist
 3. Open the cap with a twist	 3. Connect the tip of the syringe to the tubing system by turning it clockwise. Proceed according to the instructions for the device
 4. Remove the air in the syringe	

Glass syringe only:



1. Open the package
2. Screw the plunger on the syringe
3. Break the protective cover
4. Remove the protective cover
5. Remove the rubber stopper
6. Remove the air in the syringe

Manufacturer

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